

Particle Side: Inertial, Damping, ...,

Haoyi's Bouncer technique:

Freeze the position of droplet r_0 , even though it will move once you remove the arrestion.

The convergent wavefield will give the information of the force from the topography, thereby the potential for the particles.

Wave Side: tackle extremely shallow layer

Single Droplet

FREE SPACE

>> 43

Velocity Vs Phase and Memory (N=1)

>> 65 (Reference to 43)

R = 0.45 mm

Velocity Vs Phase and Memory (N=1)

TOPO

>> 56 + 61 (high resolution)

Theta = 1.00	theta = 1.10	theta = 1.20	theta = 1.30	theta = 1.35
Escape	Escape	Escape	Confined	Confined

>> 58 + 62 + 80 (Reference to 56)

R = 0.45 mm

Theta = 1.00	theta = 1.10	theta = 1.20	theta = 1.30	theta = 1.35
Escape	Escape	Escape	Escape	Confined

Collective Dynamics

PARTICLE CONFINEMENT

>> 52

theta = 1.00	theta = 1.10	theta = 1.20	theta = 1.30	theta = 1.35
Wave-like statistics	Wave-like statistics	Wave-like statistics	Parabolic	Parabolic

>> 68

R = 0.45 mm

theta = 1.20

Wave-like Statistics: REFERENCE

TOPO

>> 55

Theta = 1.00	theta = 1.10	theta = 1.20	theta = 1.30	theta = 1.35
Escape	Escape	Escape	Gaussian	Gaussian

NO WAVELIKE STATISTICS AT ALL !!!

>> 57 (Reference to 55, with larger droplet size)

R = 0.45 mm

Theta = 1.00	theta = 1.10	theta = 1.20	theta = 1.30	theta = 1.35
Escape	Escape	Escape	Gaussian	Gaussian

NO WAVELIKE STATISTICS AT ALL !!!

>> 73

h1 = 0.1 mm + Low Memory

R = 0.45 mm

Escape + Cheerios Effect

>> 81

theta = 1.20

Escape

>> 82 (domain width x2) Blew up

>> 83 (Time divisor x2) No Data

>> 84 (domain width x2 + Time divisor x2) No Data

>> 85 (Double spatial resolution) No Data

TOPO + PARTICLE CONFINEMENT

>> 39

theta = 1.20

Wave-like Statistics

>> 40 + 41 + 42 (Reference to 39)

theta = 1.25

theta = 1.35

theta = 1.40

Parabolic Statistics

Gaussian

Exponential Decaying

>> 44 + 45 (Reference to 39)

theta = 1.20

Mem = 95%

Mem = 97%

Stronger Wave-like Signiture when increasing the memory

Model Modification

- Single Droplet

SPRING FORCE

>> 48 + 49 + 50 + 51

Effective Radius = 0.8

Mem = 99.5%

Mem = 98%

Mem = 95%

Mem = 90%

Spring Provides Good Confinement

CHANGING c4

>> 88 + 89

c4 = 0.01:0.05:0.62

c4 > 0.41 Provides Good Confinement

- Collective Dynamics

SPRING FORCE + TOPO

>> 54

Effective Radius = 0.8

Mem = 90%

Mem = 95%

Mem = 98%

Mem = 99.5%

theta = 1.00

theta = 1.10

theta = 1.20

theta = 1.30

theta = 1.35

Wave-like Statistics for high memory and low impact phase (< 1.20)

SCALED Damping c4 + TOPO

>> 46

Effective Radius = 0.8, Scaling = 3

Wave-like Statistics + a lift at c4 discontinuity

>> 69

R = 0.45 mm

Effective Radius = 0.8, Scaling = 3

Wave-like Statistics + a lift at c4 discontinuity

>> 70

R = 0.45 mm

Effective Radius = 0.8 | 0.9 | 0.95 | 0.99 |

Wave-like + a lift | Wave-like + a lift | No Wave | No Wave

>> 71 + 72

R = 0.45 mm

Effective Radius = 3:1:8

Scaled damping factor = 1.0,

Not Confined !

CHANGING c4 + TOPO

>> 91 + 92 + 93

c4 = 0.31	0.41	0.62
Parabolic	Linear	Exponent (Half Wavelength Structure)

CHANGING c4 + PARTICLE CORRAL

>> 95

Large c4 is unrealistic

WAVEFIELD MASKING + TOPO

>> 86 + 87 + 88

ALL ESCAPE

PRESSURE (SPRING)

>> 63 + 64

Not a good implementation